

VRAAG / QUESTION 5

'n Eenvoudige slinger het 'n lengte ℓ en massa m . Die spankrag in die tou is T .

A simple pendulum has a length ℓ and mass m . The tension in the rope is T .



- (a) Gee die booglengte s vir klein uitwykings. / Give the arc length s for small deviations.

$$s = \ell \theta \quad \theta = \frac{s}{\ell} \quad s = \ell \theta$$

(1)

- (b) Toon aan dat die tangensiale versnelling langs die boog gegee word deur:

Show that the tangential acceleration along the arc is:

$$a_t = \ell \frac{d^2\theta}{dt^2}$$

$$a_t = r\alpha$$

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$$\text{maar } \alpha = \frac{d\omega}{dt} = \frac{d}{dt} \left(\frac{d\theta}{dt} \right)$$

$$a_t = \frac{dr}{dt} = \frac{d}{dt} \left(\ell \frac{d\theta}{dt} \right) \quad \text{want } \frac{dr}{dt} = \frac{d}{dt} \left(\ell \frac{d\theta}{dt} \right) = \ell \frac{d^2\theta}{dt^2}$$

(4)

- (c) Gee die tangensiale (of herstel) krag F_t . / Give the tangential (or restoring) force F_t .

$$F_t = -mg \sin \theta$$

(1)

- (d) Herlei 'n uitdrukking i.t.v. ℓ , g en θ wat 'n EHB beskryf.

Derive an equation in terms of ℓ , g and θ which describes a SHM.

$$\text{v. EHB moet } \propto x$$

$$F_t = -mg \sin \theta = -m \left(\ell \frac{d^2\theta}{dt^2} \right)$$

$$\text{O. } \theta \text{ klein is } \sin \theta \approx \theta$$

$$-g\theta = \ell \frac{d^2\theta}{dt^2}$$

(4)